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HOW HARD IS IT TO IDENTIFY AI-GENERATED MEME IDEAS?

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CODE & DATA
github.com/shineef/Meme-Identification

PIPELINE

Generate AI ideas on the same template as a human meme, then ask who can tell them apart.

1 · COLLECT
r/memes
Top + hot posts, Dec 3, 2025 – Mar 7, 2026: **2,735** English image memes

2 · EXTRACT
Qwen3.8-27B
One-sentence description of the template + a JSON list of the text the creator added, panel by panel

3a · HUMAN IDEA
The creator's original panel text

3b · AI IDEAS
gpt-oss-120b · Llama 4 Scout · Gemma 4 fill the same JSON schema

4 · PAIR
Idea pool
Same template: human vs. AI idea

5a · LLM PROMPTING
0 / 4-shot, with or without CoT

5b · SUPERVISED
LogReg · fastText · XLM-R · Gemma 4 + LoRA

01 THE TASK

Memes pair short text with an image and its social context, a hard case for detectors built for long-form text. Asking image models for whole memes would turn the task into spotting watermarks and image artifacts, so we isolate the **idea**: the text a creator adds to a template.

TASK
Given a template and its text schema: was this idea written by a human or an AI?

Extraction quality — 200 items · mean 4.70 · median 5

5 · perfect	76.5%
4 · mostly right	18.5%
3 · partial	4.0%
2 · mostly wrong	0.5%
1 · unusable	0.5%

02 FROM MEME TO AI IDEA

The generator gets exactly what the human creator had: the template and its text boxes. So the two ideas differ in the idea, not the format.

STEP A · EXTRACT
Qwen3.8-27B, reading the meme image

- **Template:** one sentence on the visual template only
- Keeps fixed, iconic text such as catchphrases and sign labels; drops the creator's added text
- **Human idea:** the creator's strings as a JSON list, one per text box, empty if a box is silent

STEP B · GENERATE
gpt-oss-120b · Llama 4 Scout · Gemma 4, text only

- **Input:** the template description + an empty JSON list with one slot per text box
- Asked for a meme so good that no one can tell a person did not write it
- At most 10 words per box; no strings copied from the template description
- **Output:** only the raw JSON list

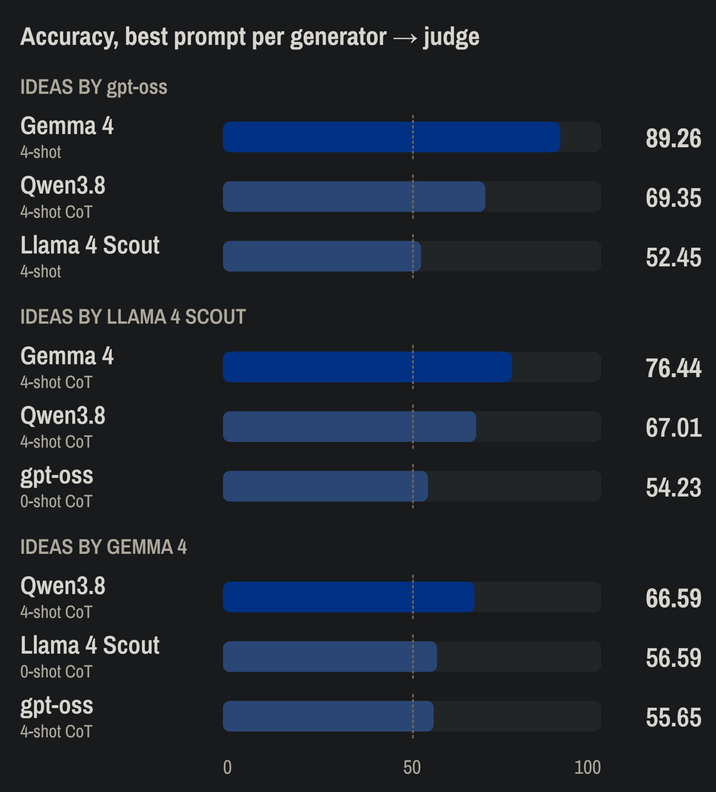
```
template   "one sentence, fixed text kept"
human      ["…", "…"]
AI         ["…", "…"] ← same slots
```

All generators: temperature 0.1, top-p 1.0, seed 0.

03 PROMPTED LLM JUDGES

Cross-evaluation: one model generates ideas and the others judge them, so no model grades its own output. Qwen3.8, our extractor, only judges. 4 prompt settings × 5 seeds.

Accuracy, best prompt per generator → judge



----- chance (50) · dark bar = best judge for that generator · mean of 5 seeds

- **Gemma 4 is the best judge:** 89.26 on gpt-oss ideas, about 20 points ahead of the next model.
- **Smaller beat bigger:** Gemma 4 (31B) and Qwen3.8 (27B) outscore the MoE models Llama 4 Scout (109B, 17B active) and gpt-oss (117B, 5.1B active).
- **gpt-oss over-calls "AI":** recall 94–96, precision 52–53.
- **The prompt matters:** 4-shot CoT wins 5 of 9 pairs and plain 0-shot never wins. Gemma 4 rises from 73.75 (0-shot) to 89.26 (4-shot).
- **Llama 4 Scout writes more human-like ideas** than gpt-oss: they are harder to catch.

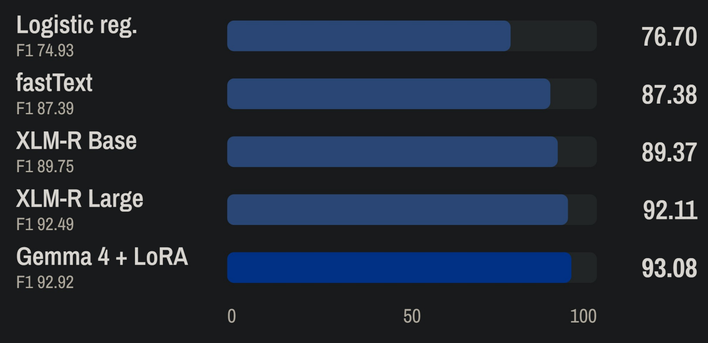
What the CoT prompt tells judges to check

AI SIGNS	HUMAN SIGNS
Boring, overly safe or sterilized tone	Edgy, risky or NSFW
Leans on the template; extra information	Puns and double meanings
Illogical, or unrelated to the template's fixed text	Niche vocabulary, specific cultural references

04 SUPERVISED CLASSIFIERS

Train 1,000 · dev 500 · test 3,970 ideas, half human, AI ideas from gpt-oss and Llama 4 Scout. Each meme stays in one split.

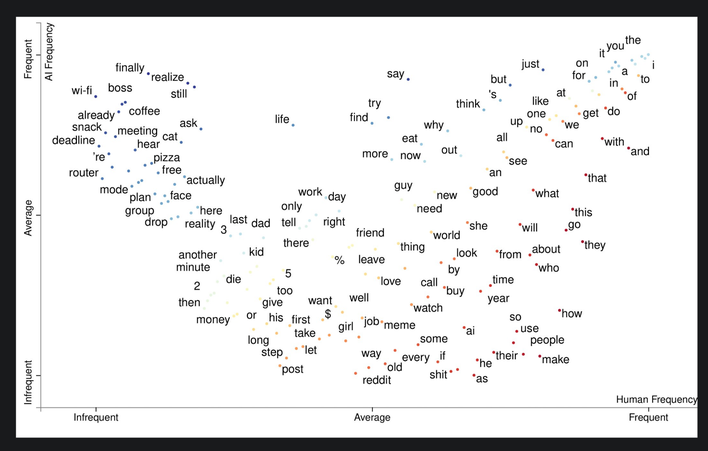
Test accuracy (F1 below each model)



Even logistic regression clearly beats the weaker prompted LLMs. fastText reaches 87.38 from just 1,000 examples, so the signal is largely lexical.

05 WHAT GIVES AI AWAY?

A clear pattern: AI ideas default to safe, "lowest common denominator" humor, mostly workplace and tech complaints. Human ideas lean on function words and general verbs: more varied syntax, no single topic.



Scattertext over lemmatized training ideas. Upper left = frequent in AI ideas, rare in human ones.

TOP AI TERMS
finally · realize · still · wi-fi · say · coffee · boss · already · snack · meeting · just

TOP HUMAN TERMS
and · they · how · make · go · this · that · people · with · use · so

Predictable, low-variation output is itself a signature of machine authorship.

TAKEAWAY

Without fine-tuning, LLMs are poor-to-fair judges of AI meme ideas, while supervised models reach 93 accuracy and F1. Today's AI humor leans on predictable work-and-tech tropes, so detection is easy, for now.

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